



TN CHENNAI DISTRICT STATEBOARD QUESTION PAPER 2024 - 2025

Follow For
More Updates





*

COMMON QUARTERLY EXAM - 2024**Standard - XII****Time : 3.00 hrs****BIOLOGY****Marks: 70****Part - I Bio-Botany****Marks: 35****Section - I****I. Answer all the questions.****8×1=8**

- 1) Size of Pollen grain in myosotis _____
a) 10 b) 2000 c) 20 d) 200
- 2) Solidifying agent used in plant tissue culture in
a) Nicotinic acid b) Cobaltous chloride c) EDTA d) Agar
- 3) What is the phenotype of wheat kernel colour for the genotype: R1R2r2r2
a) Dark red b) medium dark red
c) medium red d) Light red
- 4) Which of the given plant produces cardiac glycosides
a) Calotropis b) Acacia c) Nepenthy d) Utricularia
- 5) Match the following and find the correct answer
i) Pentasomy - A) $2n-2$
ii) Double monosomy - B) $-2n+1$
iii) Nullisomy - C) $2n-1-1$
iv) Trisomy - D) $2n+3$
a) i-C, ii-D, iii-B, iv-A b) i-B, ii-C, iii-D, iv-A
c) i-C, ii-B, iii-A, iv-D d) i-D, ii-C, iii-A, iv-B
- 6) In which techniques Ethidium Bromide is used _____
a) Southern Blotting techniques b) Western Blotting techniques
c) Polymerase chain Reaction d) Agrose gel Electroporosis
- 7) If haploid number in a cell is 18. The double monosomic and trisomic number will be _____
a) 35 and 37 b) 34 and 35
c) 37 and 35 d) 17 and 19

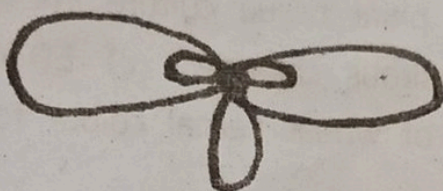
- 8) Ecology is the study of an individual species is called _____
- i) community Ecology ii) Avtecology
iii) species ecology iv) synecology
- a) i only b) ii only c) i and iv only d) ii and iii only

Section - II

II. Answer any four of the following questions:-

 $4 \times 2 = 8$

- 9) Give the names of the scientists who rediscovered mendelism.
- 10) What do you mean "EMBRYOIDS"
- 11) Draw this diagram and label the parts.



- 12) Mention the names of the enzymes required for Genetic Engineering.
- 13) What is meant by a "Cis configuration"
- 14) What is mellito phily?

Section - III

III. Answer any three of the following questions:

Q.No. 19 is compulsory:

 $3 \times 3 = 9$

- 15) What is genetic mapping? Write its uses.
- 16) Give an account on "Cryopreservation"?
- 17) Differentiate Grafting and Layering
- 18) What are multiple alleles?
- 19) Draw the flowchart for critical production.

Section - IV

IV. Answer all the questions:-

$$2 \times 5 = 10$$

- 20) a) Bring out the inheritance of chloroplast gene with an example.

[or]

- 21) a) Compare the various types of Blotting techniques.

[or]

- b) Write the applications of Plant Tissue culture.



(3)

XII BIOLOGY

Part - II Bio-Zoology

Marks: 35

Section - I

I. Choose the correct answer from the options given below and write the answer with option code:- $8 \times 1 = 8$

- 1) The Androgen Binding protein (ABP) is produced by
 - a) Sertoli cells
 - b) Leydig cells
 - c) Pituitary gland.
 - d) Hypothalamus
- 2) The condition where the women do not have functional uterus is.
 - a) Down's syndrome
 - b) Mayer-Rokitansky syndrome
 - c) Patau's syndrome
 - d) Tunner's syndrome
- 3) The term Biogenesis was coined by
 - a) Oparin
 - b) Thomas Huxley
 - c) Haldane
 - d) Henry Bastian
- 4) The total number of nitrogenous bases in human genome is estimated to be about
 - a) 3.5 million
 - b) 3.0 billion
 - c) 35,000
 - d) 35 million
- 5) Which period was called "Age of fishes"?
 - a) Permian
 - b) Triassic
 - c) Devonian
 - d) Ordovician
- 6) The age of fossils can be determined by
 - a) Electron microscope
 - b) Carbon dating
 - c) Weighing the fossils
 - d) Analysis of bones
- 7) The Athlete's foot disease in human is called
 - a) Bacteria
 - b) Fungus
 - c) Virus
 - d) Protozoan
- 8) Spread the cancerous cells to distant sites is termed as
 - a) Metastasis
 - b) Oncogenes
 - c) Protooncogenes
 - d) Maligrant neoplasm



SATGURU STUDY CENTRE

NEVER STOP LEARNING. NEVER STOP GROWING...
17/8, Poondi Thangammal St, Nagoor Garden, Tondiarpet, Chennai, Tamil Nadu 600081
An MSME-Certified Learning Centre



(4)

XII BIOLOGY

Section - II

II. Answer any four questions:-

4×2=8

- 9) Who are called as secretors?
- 10) Draw a neat labelled diagram of Human ovum.
- 11) What are the risk factors for cervical cancer?
- 12) Write the scientific name of the filarial worm that causes filariasis. and how is this disease transmitted?
- 13) What are Coprolites?
- 14) List out the major gases seems to be found in the primitive earth.

Section - III

III. Answer any three questions. Q.No.19 compulsory:

3×3=9

- 15) What is colostrum? Write its significance.
- 16) List the common withdrawal symptoms of drugs and alcohol abuse
- 17) Mention the symptoms of Phenylketonuria
- 18) State any three goals of Human Genome Project.
- 19) Write a note on Amoebiasis.

Section - IV

IV. Answer the following questions in detail:

2×5=10

- 20) a) Write the different kinds of syngamy prevalent among living organisms.

[or]

- b) Explain the structure of Immunoglobulin with a suitable diagram.

- 21) a) What are the applications of D.N.A. finger printing technique.

[or]

- b) How does Hardy-Weinberg's expression. ($P^2 + 2pq + q^2 = 1$) explain that genetic equilibrium is maintained in a population? List any four factors that can disturb the genetic equilibrium.
